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Low Mean Arterial Pressure During Cardiopulmonary Bypass and the Risk of Acute Kidney Injury: A Propensity Score Matched Observational Study

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Abstract

Introduction: Low mean arterial pressure (MAP) periods occur frequently during cardiopulmonary bypass (CPB), and their management remains controversial. Our aim was to correlate MAP during CPB with the occurrence of post-operative acute kidney injury (AKI), considering two different parameters: consecutive and cumulative low MAP periods.

Methods: Single-centre observational retrospective study including 250 patients submitted to non-emergent aortic valve replacement, with tepid to mild hypothermia (not below 32°C). The primary outcome was the occurrence of AKI. A propensity scored matching of 43 patients was used to adjust both populations (AKI and No AKI). MAP measures were automatically and continuously recorded during CPB. Low MAP periods were analysed employing two parameters: consecutive and the cumulative sum of time.

Results: Patients who experienced at least 5 min with MAP <50 mmHg had an increased risk of post-operative AKI (OR infinity; 95% CI, 1.47 to infinity; *P* = .026). The risk is also significant with MAP <40 mmHg (OR 2.78; 95% CI 1.1–6.9; = .044) and <30 mmHg (OR 3.36; 95% CI 1.2–9.2; *P* = .029). Post-operative AKI was associated with cumulative and consecutive periods of low MAP. Patients with periods of low MAP had higher levels of post-operative creatinine and reduced glomerular filtration rate (GFR). Patients with AKI had prolonged endotracheal ventilation time, and ICU and ward lengths of stay.

Conclusion: Low MAP periods during CPB are associated with an increased occurrence of post-operative AKI, leading to 1) higher creatinine levels; 2) decreased GFR and 3) longer ICU and ward lengths of stay. Both consecutive and cumulative periods of low MAP are associated with an increased risk of AKI. MAP appears to be an important contributor to post-operative AKI and should be carefully managed during CPB. Further studies must address if MAP variations lead to definitive and long-term consequences.

Keywords

mean arterial pressure, cardiopulmonary bypass, acute kidney injury, goal-direct perfusion, cardiac surgery

Introduction

Cardiopulmonary bypass (CPB) during cardiac surgery is a risk factor for acute kidney injury (AKI). New-onset post-operative AKI has an estimated incidence of up to 40% and is associated with increases in hospital length of stay,[1] morbidity and both early and late mortality.[3–5] Post-operative AKI is also associated with an increase in costs that can reach 37 to 104%, compared to patients who do not develop this complication.[2,4]

Technological advances in perfusion science over the past years have significantly increased the safety of CPB, leading to decreases in complication rates and overall mortality rates.[6] Emphasis is now on reducing morbidity, with the

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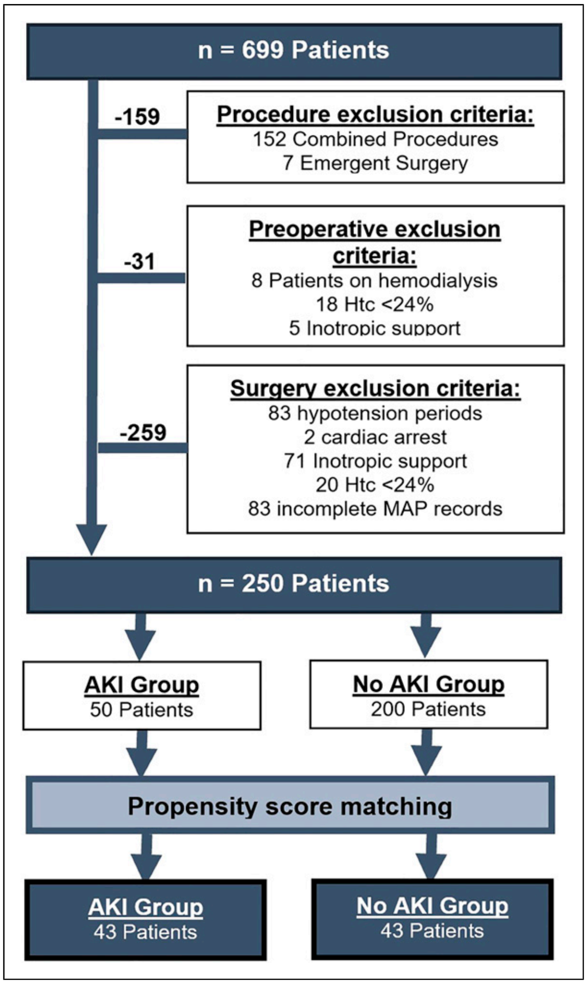

Figure 1. Flowchart of patients included in the study.

developments focussed on reducing the effects of CPB on other organs than the heart and the lungs, such as the kidneys. The occurrence of severe hypotension during CPB, with mean arterial pressure (MAP) below 55–60 mmHg, is relatively frequent[7] and is associated with an increased risk of AKI.[8,9]

During CPB, a renal oxygen supply/ demand mismatch occurs, caused partly by renal vasoconstriction.[10] Although perfusion flow rates to maintain MAP of 50 to 75 mmHg are considered adequate to preserve systemic oxygen delivery,[10] the kidney reduces the glomerular filtration rate (GFR) when MAP falls below 80 mmHg.[2] Thus, hypotension periods during CPB impair renal oxygen delivery, increasing the risk of AKI.[10] In 2015, Magruder et al.[11] described that periods longer than 15 min with a MAP below 60 mmHg were an independent risk factor for AKI.[11] Another well-known independent factor for AKI is haemodilution.[12] Anaemia has been shown to increase the risk of renal failure,[13] especially with a nadir Hct of 24%.[2,6,7]

Despite the technological and scientific advances associated with cardiac surgery, many authors suggest that flow and MAP adjustment during CPB are not based on evidence[14] but on historical[15] or even arbitrary[16] practices, which emphasizes the importance of research on this subject. There is no general recommendation for an adequate MAP to avoid AKI.[8] Reviewing the literature, we highlight the scarcity of studies that use continuous MAP recordings. Additionally, to our knowledge, no studies have been published correlating the consecutive and cumulative hypotensive periods with post-operative AKI.

In this study, our aim is to evaluate the correlation between periods of low mean arterial pressure during cardiopulmonary bypass and the incidence of post-operative acute kidney injury. Additionally, we compare the occurrence of AKI in two different scenarios: consecutive and cumulative low MAP periods.

Methods

Study Population

Our Institutional Ethical Review Board provided ethical approval for this study (Ref N.o23/18), and written informed consent was waivered. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines were followed through this study. This is a single-centre observational and retrospective study, including patients submitted to non-emergent aortic valve replacement (AVR) between January 2016 and January 2018. Exclusion criteria included 1) pre-operative conditions (patients on haemodialysis; haematocrit below nadir of 24% and patients undergoing inotropic support at the time of surgery) and 2) surgery-related conditions (presence of hypotensive periods or cardiac arrest during anaesthesia induction or right after surgery; necessity of inotropic support during surgery; haematocrit below 24% and incomplete MAP records). A total of 250 patients were included in the study (Figure 1).

Surgical Technique

Anaesthesia was induced by fentanyl and propofol, and intubation was facilitated by rocuronium. A total intravenous anaesthesia was used, maintained with propofol during CPB. The CPB circuit consisted of an Inspire 6® (Sorin Group, Italy) or Quadrox-i® (Getinge, Sweden) oxygenator, a single chamber hard-shell venous reservoir with cardiotomy filtred suction (Sorin Group, Italy) and a Sorin Adult® tubing system. A mixed priming [balanced crystalloid (polyelectrolyte) and colloid (gelatin)] of 1200cc was used to initiate CPB. Mannitol or corticoids were not used in the priming solution. Patients were anticoagulated with heparin (300 mg/Kg) before the onset of CPB, in order to achieve an activated clotting time >480 sec. Non-pulsatile roller pump was used with blood flow indexed to 2.4 L/min/m2. Surgery was performed with mild hypothermia to normothermia (target 32–36°C), controlled by a nasopharyngeal probe. Intermittent antegrade cold blood cardioplegia was used during induction and warm for reperfusion, diluted with 4 to 1 hyperkaliaemic crystalloid solution. Heparin was reversed with protamine. Blood glucose levels were maintained below 250 mg/dL. Minimal allowable haematocrit during CPB was 24%. In case of hypotension during CPB, the first-line treatment was to increase pump output to a maximum of 2.6 L/min/m2. If hypotension persisted, with the maximum pump output, we initiated vasopressors, firstly with single bolus of norepinephrine and/ or epinephrine, and then initiating an infusion of norepinephrine and/ or epinephrine. Patients with the necessity of initiating a vasopressor infusion were excluded from this study (Figure 1). All patients were transferred to the intensive care unit (ICU) after surgery.

Data Collection

Baseline data included clinical characteristics, past medical history and comorbidities and pre-operative serum creatinine and haematocrit. Pre- and perioperative data were retrieved retrospectively from the clinical files from our department and from registries from the national electronic healthcare database. Details from CPB were collected from the medical records completed during the surgery.

MAP values were automatically recorded in all patients during the surgery through the arterial line placed in the right radial artery, and stored by our software Anaesthesia Manager® (Picis Clinical Solutions, United States). MAP values were collected with continuous measurements during CPB.

To analyse the correlation between MAP and acute kidney injury, we have performed two different analyses considering: 1) the cumulative time with MAP under 50, 40 or 30 mmHg (the sum of all periods of MAP under the defined value) and 2) the longest consecutive period with MAP under 50, 40 or 30 mmHg. The two analyses were performed to understand if the sum of periods of low MAP had the same impact on post-operative AKI as an equal consecutive measurement of low MAP.

Baseline serum creatinine and haematocrit were obtained from standard pre-operative bloodwork. Post-operative serum creatinine and haematocrit were measured 1 h and 24 h after surgery in the ICU.

Definitions and Outcomes Measures

The primary outcome was the occurrence of post-operative AKI. We used the Acute Kidney Injury Network (AKIN) criteria[17,18] to classify AKI.

Statistical Analysis

No statistical power calculation was conducted prior to the study, and the sample size was based on the available data. Continuous variables are expressed as median with interquartile range (IQR) and were analysed using the Mann–Whitney *U* test. Categorical variables are reported in frequencies or percentages. We compared categorical variables with the chi-square test or Fisher exact test, using the Baptista–Pike method when one of the values from the contingency table was 0. All reported *P*-values are 2-sided, and *P*-values of <.05 were considered statistically significant. To investigate the influence of hypotensive periods in post-operative AKI in both groups, while reducing the effect of confounding, a 1:1 propensity score match (PSM) was performed. Patients were matched using caliper matching method without replacement with a caliper width of .1 SD of the logit of propensity score. We have included in the PSM 12 variables: age, sex, hypertension, diabetes mellitus, atrial fibrillation, chronic kidney disease, ischaemic cardiopathy, previous cardiac surgery, left-ventricular function, EuroscoreII and cardiopulmonary bypass and aortic cross-clamp times. The balance of the covariates was tested using standardized mean difference (SMD), achieving an adequate matching with SMD <.1. No adjustments for significant level of P were made for multiple comparisons. GraphPad Prism®, version 6, and XLSTAT® for Macintosh were used for statistical analysis.

Results

Patient's Characteristics

A total of 250 patients were enrolled in this study. Exclusion criteria are detailed in Figure 1. After propensity score matching, 43 patients were included in each group (AKI and No AKI) (Figure 1). The baseline characteristics for both the unmatched and matched populations are detailed in Table 1. Considering all 250 patients, 20% (50 patients) experienced post-operative AKI, with 18% (45 patients) with AKIN 1, .4% (1 patients) AKIN 2 and 1.6% (4 patients) with AKIN 3 criteria. After matching, no differences were observed in age, sex and comorbidities such as chronic kidney disease or diabetes mellitus. Perioperative characteristics are summarized in Table 2.

Mean Arterial Pressure and Post-operative outcomes

The incidence of post-operative AKI was higher in patients with at least 5 minutes with MAP under 50 mmHg (OR infinity; 95% CI, 1.47 to infinity; *P* = .026) [Figure 2(A)]. The incidence of AKI was also higher when patients had at least 5 min with lower values of MAP, such as under 40 mmHg (OR 2.78; 95% CI 1.1–6.9; *P* = .044) (Figure 2(A)) and under 30 mmHg (OR 3.36; 95% CI 1.2–9.2; *P* = .029) [Figure 2(A)]. Patients with periods of lower MAP had a higher percentage of more severe AKI (Figure 2(B)).

Considering the patients with MAP under 50 mmHg, the risk was progressively higher with higher cumulative and consecutive periods of hypotension [Figure 2(C)]. No significant differences were observed between consecutive and cumulative periods. Patients with 10 min of cumulative MAP below 50 mmHg had an increased risk of post-operative AKI (OR 4.03; 95% CI 1.4–11.6; *P* = .014), similar to those who

Table 1. Demographic data.

	Unmatched			Matched		
Variable, N (%)	AKI (*n*=50)	No AKI (*n*=200)	*P*-value	AKI (*n*=43)	No AKI (*n*=43)	*P*-value
Age (years) median; IQR	76; 71-80	73; 67-77.8.	*0.02	75; 69-7869-78	75; 69-79	.97
Male	30 (60)	109 (54.5)	.53	27 (62.8)	28 (65.1)	1.00
Hypertension	45 (90)	182 (91)	.79	40 (93)	40 (93)	1.00
Diabetes mellitus	16 (32)	65 (32.5)	1.00	14 (32.6)	16 (37.2)	.82
Dyslipidemia	36 (72)	144 (72)	1.00	32 (74.4)	32 (74.4)	1.00
Atrial fibrillation	13 (26)	41 (20.5)	.44	10 (23.3)	13 (30.2)	.63
Chronic kidney disease	10 (20)	14 (7)	*0.01	6 (13.9)	6 (13.9)	1.00
Cerebrovascular disease	6 (12)	9 (4.5)	.09	6 (13.9)	3 (7)	.48
Chronic lung disease	3 (6)	20 (10)	.58	2 (4.7)	3 (7)	1.00
Ischaemic cardiopathy	8 (16)	19 (9.5)	.20	6 (13.9)	8 (18.6)	.77
Previous cardiac surgery	0	4 (2)	.59	0	1 (2.3)	1.00
Left ventricular function preserved (>50%) moderate (31–50%) poor LV (21–30%) very poor (<20%)	43 (86) 6 (12)1(2) 0	180 (90) 19 (9.5) 1 (.5)0	.45 .60 .36 –	40 (93) 3 (7)0 0	38 (88.4) 5 (11.6) 0 0	.71 .71 – –
NYHA I	3 (6)	7 (3.5)	.42	2 (4.7)	1 (2.3)	1.00
NYHA II	35 (70)	150 (75)	.48	32 (74.4)	36 (83.7)	.43
NYHA III	12 (24)	41 (20.5)	.57	9 (20.9)	6 (14)	.57
NYHA IV	0	2 (1)	1.00	0	0	1.00
EuroSCORE II	1.33; 2.03	1.01; .73	.12	.91; .74	.96; .56	.48

IQR: interquartile range; LV: left ventricular; NYHA: New York Heart Association.
Chronic kidney disease: kidney damage, as defined by structural or functional abnormalities of the kidney, with or without decreased GFR, or GFR of less than 60 mL/min/1.73 m^2, for at least 3 months.
Cerebrovascular disease: neurological deficits due to vascular insufficiency or occlusion, or haemorrhage, including the previous diagnosis of stroke, carotid stenosis, vertebral stenosis and intracranial stenosis, aneurysms and vascular malformations.
Chronic lung disease: diagnosis of chronic obstructive pulmonary disease, asthma, or other lung disease, with the necessity of long-term use of bronchodilators and/ or steroids.
Ischaemic cardiomyopathy: significantly impaired left ventricular function that results from coronary artery disease.

Table 2. Surgery and post-operative data.

	Unmatched			Matched		
Variable, N (%)	AKI (*n*=50)	No AKI (*n*=200)	*P*-value	AKI (*n*=43)	No AKI (*n*=43)	*P*-value
Aortic stenosis	47 (94)	190 (95)	.727	40 (93)	42 (97.7)	.727
Aortic insufficiency	3 (6)	10 (5)	.727	3 (7)	1 (2.3)	.727
Endocarditis	0	0	–	0	0	–
Aortic prothesis dysfunction	0	0	–	0	0	–
Biological prothesis	47 (94)	186 (93)	1.000	41 (95.3)	42 (97.7)	1.000
Mechanical prothesis	3 (6)	14 (7)	1.000	2 (4.7)	1 (2.3)	1.000
Morrow miectomy	5 (10)	32 (16)	.375	4 (9.3)	4 (9.3)	1.000
Elective surgery	39 (78)	179 (89.5)	.055	33 (76.7)	36 (83.7)	.589
Urgent surgery	11 (22)	21 (10.5)	.055	10 (23.3)	7 (16.3)	.589
Extracorporeal circulation (min)	50; 21	50; 28.8	.234	50; 21	50; 18	.477
Aortic cross-clamp time (min)	34; 22.4	33; 18	.366	32; 25	36; 19	.689
Ventilation (hours)	7.8; 7	5; 3	**0.005	8; 7	5; 3.5	.073
Chest tube drenaige 24 h (ml)	400; 762	300; 300	.207	400; 800	400; 400	.635
ICU stay (hours)	84; 60	48, 48	**** <.0001	72; 72	48; 48	*** .0004
Ward stay (days)	7; 3.8	6; 2	**** <.0001	7; 3.5	6; 2	** .002
Discharge, n (%)	41 (85)	185 (93)	.154	34 (83)	39 (95)	.345
Other hospital	7 (15)	14 (7)		7 (17)	4 (5)	
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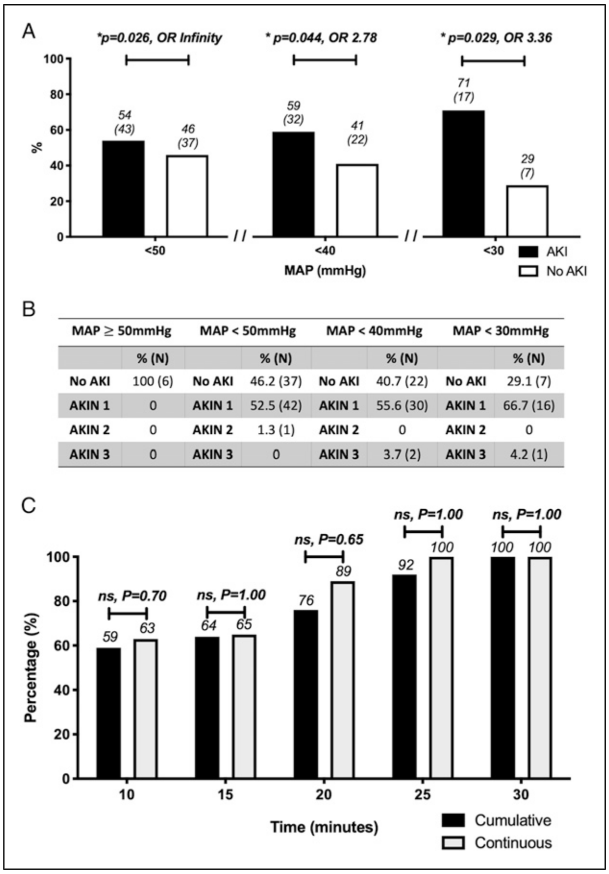
Continuous variables are presented as median; IQR. IQR: interquartile range; AKI: acute kidney injury; ICU: intensive care unit



MAP ≥ 50mmHg		MAP < 50mmHg		MAP < 40mmHg		MAP < 30mmHg	
	% (N)		% (N)		% (N)		% (N)
No AKI	100 (6)	No AKI	46.2 (37)	No AKI	40.7 (22)	No AKI	29.1 (7)
AKIN 1	0	AKIN 1	52.5 (42)	AKIN 1	55.6 (30)	AKIN 1	66.7 (16)
AKIN 2	0	AKIN 2	1.3 (1)	AKIN 2	0	AKIN 2	0
AKIN 3	0	AKIN 3	0	AKIN 3	3.7 (2)	AKIN 3	4.2 (1)

Figure 2. Proportion of matched patients developing post-operative AKI with periods of 5 min of cumulative MAP under 30, 40 and 50 mmHg. **(A)** Proportion of matched patients with AKI, comparing the occurrence of AKI in patients without hypotension with patients with periods of MAP under 50, 40 and 30 mmHg. **(B)** Percentage of patients with or without AKIs (detailed by AKIN stages) considering patients without hypotension or with periods of low MAP. **(C)** Proportion of patients with post-operative AKI with cumulative or consecutive periods of MAP below 50 mmHg. OR: odds ratio; MAP: mean arterial pressure; AKI: acute kidney injury; AKIN: Acute Kidney Injury Network Criteria. The following symbols were used in figures to indicate statistical significance: $P < .05$ (*); $P < .01$ (**); $P < .001$ (***); $P < .0001$ (****).

experienced 10 min of consecutive MAP<50 mmHg (OR 3.2; 95% CI 1.3–7.8; P = .016). For longer periods, such as 20 min, the risk of post-operative AKI seemed to be higher with consecutive periods (OR 9.6; 95% CI 1.14–80.6; P = .029) than with cumulative periods (OR 4.9; 95% CI 1.7–13.9; P = .004), although the proportion of patients was similar in our population.

Patients with post-operative AKI had an increased ICU length of stay (72; IQR 72) hours vs (48; IQR 48) hours, P = .0004] and ward stay (7; IQR 3.5 days vs 6 IQR 2 days; P = .002) (Table 2). Similarly, the occurrence of AKI was associated with prolonged ventilation time, although it is not statistically significant (Table 2).

Mean Arterial Pressure and Glomerular Filtration Rate

As periods of lower MAP were associated with an increased risk of AKI, we evaluated if they were associated with differences in post-operative creatinine levels and glomerular filtration rate (GFR).

Pre-operative creatinine levels were not statistically different when comparing patients without hypotension and patients with periods of hypotension (Figure 3). Patients with hypotensive periods had a significant increase in creatinine levels in the first 24 h after the surgery (Figure 3). However, no significant differences were observed in 1 h and 24 h post-operative creatinine levels in patients who maintained MAP above 50 mmHg (Figure 3).

We also examined the association between hypotension and post-operative GFR (Figure 4). In patients without hypotensive periods, mean GFR 24 h after surgery was 64.9; IQR 40.4 mL/min, with no significant differences compared to pre-operative values (Figure 4). However, a statistically significant difference was observed in GFR at 24 h after surgery in patients with MAP periods below 50 mmHg (52.9; IQR 30.5 mL/min; P <.0001), 40 mmHg (53.8; IQR 30.7 mL/min; P <.0001) and 30 mmHg (61; IQR 32.4 mL/min; P = .0001). No significant differences were observed between pre-operative values of all groups.

Discussion

In the current study, we evaluated the association between hypotension during CPB and the occurrence of post-operative AKI. The main findings were that periods of MAP under 50 mmHg are associated with a higher incidence of AKI, with increased post-operative creatinine levels and reduced glomerular filtration rate. Both cumulative and consecutive periods of low MAP contribute similarly to post-operative AKI. Our results suggest that MAP appears to be an essential variable to cautiously manage during CPB. We excluded patients with Htc of less than 24% and maintained a pump index flow of ≥2.4 L/min/m2, and thus assumed an adequate oxygen delivery, to guarantee that our results were not associated with a dysfunctional oxygen delivery, although this was not monitored.

It is well established that CPB is a risk factor for AKI. CPB impairs the renal oxygen supply/ demand relationship, leading to renal injury.[10] Both renal medulla and cortex are highly susceptible to damage during cardiac surgery, especially due to the hypoperfusion during the rewarming phase.[19] Newland et al. have demonstrated that there is a relation between oxygen delivery during CPB and the incidence of post-operative AKI,[20] such that the risk of post-operative AKI is reduced when oxygen delivery remains above 280–300 mL/min/m2.[21,22]

MAP variability has an important role in the aetiology of AKI[3] as changes in blood flow may damage the kidney.

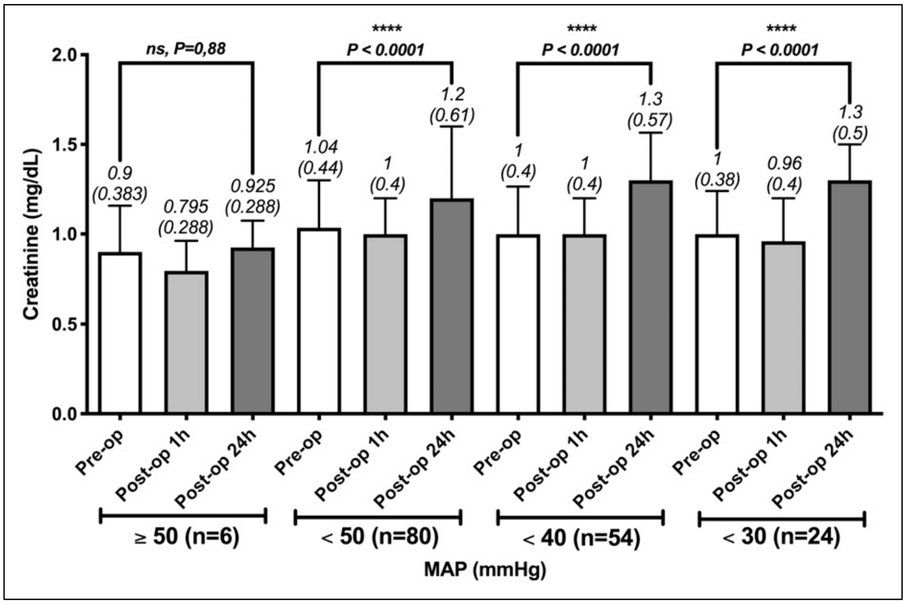


Figure 3. Mean basal creatinine levels and 1 h and 24 h after surgery in patients with periods of 5 min of cumulative MAP below 30, 40 and 50 mmHg, after propensity score match. MAP: mean arterial pressure, ns: non-significant. The following symbols were used in figures to indicate statistical significance: $P < .05$ (*); $P < .01$ (**); $P < .001$ (***); $P < .0001$ (****).

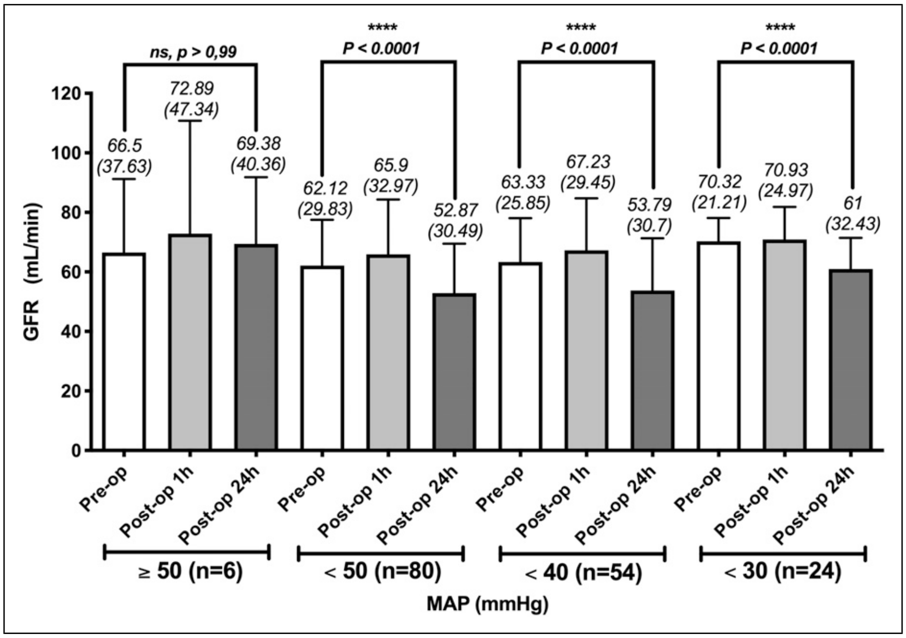


Figure 4. Mean GFR before and after surgery (1 h and 24 h) in matched patients with periods of 5 min of cumulative MAP below 30, 40 and 50 mmHg (C). GFR: glomerular filtration rate; MAP: mean arterial pressure; ns: non-significant. The following symbols were used in figures to indicate statistical significance: $P < .05$ (*); $P < .01$ (**); $P < .001$ (***); $P < .0001$ (****).

Kidneys have autoregulatory mechanisms that maintain normal glomerular function between 75 and 160 mmHg.[14] Thus, MAP under 80 mmHg leads to an important reduction in the GFR.[23] On the other hand, the restoration of MAP from 60 to 75 mmHg in patients with AKI after cardiac surgery increases GFR and improves renal oxygenation.[24]

Although renal lesions can be reversible after surgery, prolonged hypoxia or hypotension periods may lead to permanent or clinically relevant dysfunction.[19]

Several studies in the past few years have observed that MAP was not associated with the risk of AKI.[14,25,26] However, many of these studies included procedures with deeper degrees of hypothermia, big cohorts of arterial pressure (e.g. low pressure group with MAP between 39 and 70 mmHg) and technical limitations with values manually recorded, potentially incomplete (probably missing additional covariates) and biased. In our study, we present a MAP analysis based on continuous and automatically recorded values from the patient arterial line. To our knowledge, it is one of the first studies to correlate automatically continuous MAP measures and the risk of post-operative AKI in normothermic or mild hypothermic patients.

In the current study, we show that just 5 min with MAP under 50 mmHg increases the risk of AKI. The CPB and aortic clamping times presented here (Table 2) are shorter than other previously published series, thus reinforcing that MAP is an essential physiological factor during CPB, even for short periods.

Additionally, our results indicate that both consecutive and cumulative periods of low MAP increase the risk of post-operative AKI. Both consecutive and cumulative time groups had a similar proportion of patients with post-operative AKI, without any statistically significant difference. Our findings may have an important clinical impact, suggesting that the cumulative effect of multiple separate periods of brief hypotension may be just as damaging as a prolonged and continuous individual episode of hypotension. Our results are supported by previous studies that suggest that regardless of having higher values of MAP during the remainder of the procedure, isolated periods of hypotension during CPB are associated with AKI.[27] Unfortunately, the population of our study does not have enough individuals with longer low MAP periods to perform a more detailed and consistent analysis.

Accordingly, patients who experienced low MAP periods also had significant increases in post-operative creatinine and significant decreases in GFR. On the other hand, creatinine levels and GFR were not significantly changed 24 h after surgery in patients without any hypotensive period during CPB. Elmiteskawy et al. have shown that even small reductions in GFR increase the risk of morbidity and death.[28] The impact of this reduction on long-term outcome remains unknown, and to our knowledge, no studies have addressed the correlation between MAP during CPB, permanent GFR reduction and long-term outcome. Future studies should address this issue, especially considering the fact that it is well established that AKI substantially increases the risk of chronic kidney disease and its progression to end-stage renal disease.[29]

We found that patients with post-operative AKI have a prolonged ventilation time, although it is not statistically significant, and an increased ICU and ward length of stay. Our results suggest a possible relationship between low periods of MAP and renal dysfunction. Thus, MAP should be cautiously managed during CPB to mitigate organ dysfunction as it has as important impact on morbidity and mortality, as well as on the increase of healthcare costs.

Limitations

There are several limitations associated with this study. It was a retrospective, single-centre observational study. As a single-centre study, our findings are related to our population and may not be extrapolated to other populations. Not all patients included initially in the study had complete MAP measures and had to be excluded. MAP values were collected through a radial arterial line that does not entirely represent central arterial and renal perfusion pressure during CPB. Our CPB machines do not have an integrated goal directed perfusion system, and an adequate oxygen delivery was assured maintaining haematocrit levels above 24% with an adequate pump output. Relatively low number of patients in some subgroups is also a limitation. Surgeries were not all performed by a single surgeon, and post-operative follow-up was performed by different medical and nurse teams at the ICU. All patients were submitted to aortic valve replacement, and other procedures can have different outcomes.

Conclusion

In summary, hypotensive periods of MAP under 50 mmHg increase the risk of AKI, leading to significantly higher post-operative creatinine levels and decreased GFR. Furthermore, the risk of AKI is increased both with cumulative and consecutive periods of hypotension. MAP might be an important contributor to post-operative morbidity, since patients with AKI have increased ventilation time and ICU and ward lengths of stay. MAP must be carefully managed during CPB to reduce the risk of AKI. However, it is not yet established whether these variations in MAP lead to definitive and long-term consequences, and further studies should be addressed to establish this correlation.

Declaration of conflicting interests

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